

Spiking Data Preprocessing Guide

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Code File Appendix

This is an appendix with the important coding files in the spike processing pipeline. There are more functions and files used, but these are the most important and the ones that you will likely make changes to based on your use case.

Neural-Data-Reading/Processing_Functions_cgg

All Data Processing

- DATA_cggAllSessionInformationConfiguration.m
- DATA_pgpAllSessionSpikeProcessing_v2.m

Parameters

- PARAMETERS_cgg_getSessionProbeInformation.m
- PARAMETERS_cgg_procFullTrialPreparation_v2.m

Neural_Data_cgg

- pgp_proc_EventFrameTrialPreparation.m
- pgp_procProcessingBehaviorOnly.m
- pgp_buildAlignFromEpoch.m

Input_Output_Information

- pgp_exportCfgToJson.m

Spike_Alignment

- `pgp_spikeTrialAlignment_v4.py`

Step 1 - Prerequisites

1. In `DATA_cggAllSessionInformationConfiguration.m` and `PARAMETERS_cgg_getSessionProbeInformation.m`, add in the session names and the probe channels and numbers for your respective experiment and monkey.
2. In `PARAMETERS_cgg_procFullTrialPreparation_v2.m` choose your alignment event and baseline parameters. Currently, this should be set to:

```
case 'Decision' Frame_Event_Selection_Data = 'SelectObject';  
Frame_Event_Selection_Location_Data = 'END';  
Frame_Event_Selection_Baseline = 'Blink';  
Frame_Event_Selection_Location_Baseline = 'START'; end  
Window_Before_Data = 1.5; Window_After_Data = 1.5; % 500 ms  
after blink in start of trial Window_Before_Baseline = 0;  
Window_After_Baseline = 0.5; % might also be something like  
below: % less likely to go into object selection during trial %  
Window_Before_Baseline = 0.15; % Window_After_Baseline = 0.4;
```

This will segment trials around the beginning of object selection and take a baseline from the start of the trial.

3. Make sure to have a python3 virtual environment with the necessary python packages to run the `pgp_spikeTrialAlignment_v4.py` script installed (numpy, pandas, scipy, check file for any other packages).
4. You will also need the sorted KS4 data from David and the `<session_name>_BHV` folders and potentially the record node folders from `DATA_Neural` (original recording files)

Step 2 - Main Code

The main code runs the master file: DATA_pgpAllSessionSpikeProcessing_v2.m

In this, 5 main scripts are executed

1. `pgp_proc_EventFrameTrialPreparation.m` which creates `trialDef` and `gameframedata` files with the recording times for each trial and event
2. `pgp_precprocessingBehaviorOnly.m` which segments trials and gets your trial variables file from `pgp_getTrialVariables`. The trial variables file contains all the info about the Trial number, trial in block number, block, object selected, selected location, etc... To modify this change the `pgp_getTrialVariables` script.
3. `pgp_buildAlignFromEpoch.m` creates an alignment file that that is used to align the spike times to each trial. This is run for both the task data and the baseline data.
4. `pgp_exportCfgToJson.m` exports information about the session from `DATA_cggAllSessionInformationConfiguration.m` and info about the segmentation parameters and other details from `PARAMETERS_cgg_procFullTrialPreparation_v2.m`. You will have to point the script to where you want this saved.
5. `pgp_spikeTrialAlignment_v4.py` is run through the matlab terminal which lines up the spike times with the Salign file generated and segments all trials and baselines and generates your spiking files. It also does some quality assurance checks to make sure that all the spike times in your aligned files are correct.

Things You Need to do to run:

1. Have your python environment running in the terminal.
2. point `json_out` to where you want the json saved to
3. point `sorted_root_base` to where the sorted spike files are. Currently they are both on TEBA and on ACCRE under a folder called `'M1VU595_ACC_PFC_CD/sorted'`

4. make sure that the arguments in the system call which runs the spike trial alignment match the parameters used for trial segmentation. e.g. make sure that `--win_before`, and `--win_after` reflect the parameters used to get trial time during the task and `--win_before_base` and `--win_after_base` have the correct before and after times from `PARAMETERS_cgg_procFullTrialPreparation_v2.m`
5. Run the script

Part 3 Code Output

The output of this will be a .mat in a folder called SpikeAligned called `<session_name>_SpikeCounts.mat`

In this is a cell array called X that has an entry for each trial with a cell that has a matrix that is `num_neurons x num_timepoints` as well as `X_base` which has the corresponding baseline. `TrialId` is simply the number of cells in X aka the number of trials. It should correspond precisely to the number of rows in the `TargetInformation.mat` file which can be found in `Session/EpochData/Decision/Target`. `Meta` contains info such as the sampling frequency, how much time there was before and after event and baseline, number of probes, and info about each unit such as what probe it belongs to which is important for sorting units into their respective areas.